

Modification of the perturbation

class_public/source/perturbations.c

pba->has_scf

```
*/
if (pba->has_scf == _TRUE_) {

    if (ppt->gauge == synchronous){
        delta_rho_scf = 1./3.*
            (1./a2*ppw->pvecback[pba->index_bg_phi_prime_scf]*y[ppw->pv->index_pt_phi_prime_scf]
            + ppw->pvecback[pba->index_bg_dV_scf]*y[ppw->pv->index_pt_phi_scf]);
        delta_p_scf = 1./3.*
            (1./a2*ppw->pvecback[pba->index_bg_phi_prime_scf]*y[ppw->pv->index_pt_phi_prime_scf]
            - ppw->pvecback[pba->index_bg_dV_scf]*y[ppw->pv->index_pt_phi_scf]);
    }
    else{
        /* equation for psi */
        psi = y[ppw->pv->index_pt_phi] - 4.5 * (a2/k/k) * ppw->rho_plus_p_shear;

        delta_rho_scf = 1./3.*
            (1./a2*ppw->pvecback[pba->index_bg_phi_prime_scf]*y[ppw->pv->index_pt_phi_prime_scf]
            + ppw->pvecback[pba->index_bg_dV_scf]*y[ppw->pv->index_pt_phi_scf]
            - 1./a2*pow(ppw->pvecback[pba->index_bg_phi_prime_scf],2)*psi);
        delta_p_scf = 1./3.*
            (1./a2*ppw->pvecback[pba->index_bg_phi_prime_scf]*y[ppw->pv->index_pt_phi_prime_scf]
            - ppw->pvecback[pba->index_bg_dV_scf]*y[ppw->pv->index_pt_phi_scf]
            - 1./a2*pow(ppw->pvecback[pba->index_bg_phi_prime_scf],2)*psi);
    }
}
```

pba->has_fld

```
/* fluid contribution */
if (pba->has_fld == _TRUE_) {

    class_call(background_w_fld(pba,a,&w_fld,&dw_over_da_fld,
                                w_prime_fld = dw_over_da_fld * a_prime_over_a * a));

    if (pba->use_ppf == _FALSE_) {
        ppw->delta_rho_fld = ppw->pvecback[pba->index_bg_phi_prime_fld]*y[ppw->pv->index_pt_phi_prime_fld]
        ppw->rho_plus_p_theta_fld = (1.+w_fld)*ppw->rho_plus_p_theta_scf
        ca2_fld = w_fld - w_prime_fld / 3. / (1.+w_fld)
        /** We must gauge transform the pressure perturbation */
        ppw->delta_p_fld = pba->cs2_fld * ppw->delta_rho_fld
    }
}
```